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(1) Research Program/Project/Study Title: Efficacy of Langkawas (<i>Alpinia galanga</i>) and Betel Nut (<i>Areca catechu</i>) Extract in Darag Native Chicken (<i>Gallus gallus domesticus</i>)	
Proponent Name:	Dr. Maryneth B. Barrios
Academic Rank:	Associate Professor V
Co-Authors (with academic rank):	Dr. Leah A. Ingalla – Asso. Prof V Dr. Joseph M. Lachica – Asso. Prof III
Campus/Satellite College/Address:	Dumarao Satellite College
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Program or Project has component (if no proceed to 2)? ☒ Yes ☐ No

Component 1 Title:	Anthelmintic Effects of Langkawas (<i>Alpinia galanga</i>) and Betel Nut (<i>Areca catechu</i>) Extract Against Helminths of Darag Native Chicken (<i>Gallus gallus domesticus</i>)
Component Leader:	Dr. Maryneth B. Barrios
Academic Rank:	Associate Professor V
Component 2 Title:	Anticoccidial Effects of Langkawas (<i>Alpinia galanga</i>) and Betel Nut (<i>Areca catechu</i>) Extract Against Coccidia of Darag Native Chicken (<i>Gallus gallus domesticus</i>)
Component Leader:	Dr. Maryneth B. Barrios
Academic Rank:	Associate Professor V
Component 3 Title:	Effect of Langkawas (<i>Alpinia galanga</i>) and Betel Nut (<i>Areca catechu</i>) Extract in Blood and Serum Biochemical Profile of Darag Native Chicken (<i>Gallus gallus domesticus</i>)
Component Leader:	Dr. Leah A. Ingalla
Academic Rank:	Associate Professor V
Component 4 Title:	Performance of Darag Native Chicken (<i>Gallus gallus domesticus</i>) Given with Langkawas (<i>Alpinia galanga</i>) and Betel Nut (<i>Areca catechu</i>) Extract
Component Leader:	Dr. Joseph Lachica
Academic Rank:	Associate Professor III

(2) Category (please check) ☐ Social Science/Education/Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/Computer/Engineering/Architecture ☐ Extension	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC
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		Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation		
Municipality/City:		Dumarao
Province:		Capiz
Region:		6
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Livestock		(8) Discipline/Field of Specialization Animal Science

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(9) Rationale

The prevalence of most parasitic diseases in poultry seems to have been reduced significantly in commercial indoor poultry production systems due to improved housing, hygiene and management. However, parasitic diseases continue to be of great importance in deep-litter and free-range commercial systems. In traditional systems throughout the world a number of parasites are widely distributed and contribute significantly to the low productivity. The most commonly mentioned parasites are *Eimeria spp.*, *Ascaridia galli* and *Heterakis gallinarum* which is mainly due to the many studies carried out on these parasites (Permin and Hansen, 1998).

Drugs and live vaccines are the control measures used recently in the poultry farms (Abbas, 2012). But development of drug resistance in strains in the field and the withdrawal period for these drugs prior to slaughter necessitate the exploration of alternative methods (Chandrakesan *et al.*, 2009). Drug resistance causes the effectiveness of therapeutic drugs to be suboptimal. The development of drug resistance in helminths to chemotherapeutical products has drawn attention to the need for alternative treatments. Anthelmintics from natural sources could play a key role in the treatment of parasitic infections (Prastowo *et al.*, 2017). Some of the plants that can be used Langkawas (*Alpinia galanga*) and Betel nut (*Areca catechu*).

Langkawas (*Alpinia galanga*) plant which is associated with family Zingiberaceae is mainly scattered in tropical areas and widely known for ethno medicine. Against fungi and bacteria rhizome extract have a maximum inhibitory effect. *Alpinia galanga* plant is used in medicine and in food preparation. *Alpinia galanga* also contains many flavonoids and a wide range of volatile oils. The rhizome of *Alpinia galanga* is used in many regions for relieving toothache, muscular swelling, rheumatism, abdominal pain. Terpinen-4-ol is one of the most active compounds of *Alpinia galanga* and this plant also contains, saponins, terpenoids, phenolics, flavonoids, carbohydrates, alkaloids, glycosides, phytosterols. Ethanolic extract of *Alpinia galanga* isolates galango flavonoid and also have several medicinal active ingredients that have the potential of antitumor, antifungal, antioxidant, hypoglycemic, gastroprotective, hypolipidemic, antiinflammatory activity (Eram *et al.*, 2019).

Areca catechu or betel nut or bunga is a species of palm grown mainly in Asian countries, like the Philippines for seed crop. Betel nut, the fruit of *Areca catechu* L, has a long medical history in Southeast Asia. Betel nut almost appears as a “snack” in various occasions in most parts of China. Clinically, betel nut can play a certain pharmacology role and was used in malaria, ascariasis, arthritis, enterozoic abdominalgia, stagnation of food, diarrhea, edema, and beriberi (Chen *et al.*, 2021). Betel nut contains chemical components such as alkaloids, arecoline, flavonoids, tannins and many others. Several studies showed the efficacy of this betel nut against intestinal worms of different species of animals including goats, cattle and chicken.

The CAPSU Darag Native Chicken Research Project has abundant supply of these two plants (Langkawas and Betel nut) hence, this study will be conducted to evaluate the potential effects of these plants extract in Darag Native Chicken.

(9-A) Project Description

This study will be conducted at the Darag Research Project at the College of Veterinary Medicine of CapSU Dumarao. Mature betel nuts and langkawas rhizome will be used in this study. This plants will be chopped, dried and subject fro ethanolic extraction at WVSU laboratory. Experimentals animals to be used will be screened for the presence of parasites before the study will be conducted. Data on anthelmintic, and anticoccidial effect of the plant extracts will be conducted at the CapSU CVM Laboratory. The blood and serum biochemical serum profile will be submitted for the laboratory analysis in Biofera Roxas. On the other hand, data on performance will be the productive performance of the experimental Darag native chicken.

(10) Objectives

General Objective: Evaluate the efficacy of Langkawas (*Alpinia galanga*) and Betel Nut (*Areca catechu*) extract in Darag Native Chicken (*Gallus gallus domesticus*)

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Specific Objectives

Component 1. Anthelmintic Effects of Langkawas (*Alpinia galanga*) and Betel Nut (*Areca catechu*) Extract Against Helminths of Darag Native Chicken (*Gallus gallus domesticus*)

- Evaluate the efficacy of Langkawas and Betel nut extract against roundworms and tapeworms of Darag native chicken.
- Differentiate the effect of the Langkawas and Betel nut extract in the percent reduction of the egg per gram counts of helminths.
- Compare the efficacy of plants extracts in the commercial dewormer.
- Find out the net value of using Langkawas and Betel nut extracts as dewormer to chicken.

Component 2. Anticoccidial Effects of Langkawas (*Alpinia galanga*) and Betel Nut (*Areca catechu*) Extract Against Coccidia of Darag Native Chicken (*Gallus gallus domesticus*)

- Evaluate the efficacy of Langkawas and Betel nut extract against coccidia of Darag native chicken.
- Differentiate the effect of the Langkawas and Betel nut extract in the percent reduction of the oocyst per gram counts of coccidia.
- Compare the efficacy of plants extracts in the commercial coccidiostat.
- Find out the net value of using Langkawas and Betel nut extracts as coccidiostat to chicken.

Component 3. Effect of Langkawas (*Alpinia galanga*) and Betel Nut (*Areca catechu*) Extract in Blood and Serum Biochemical Profile of Darag Native Chicken (*Gallus gallus domesticus*)

- Evaluate the effect of serpentina plant powder in the hematological parameters of Darag native chicken in terms of:
 - a. Red blood cell (RBC) Counts
 - b. White blood cell (WBC) Counts
- Evaluate the effect of serpentina plant powder in the serum biochemical parameters of Darag native chicken in terms of:
 - a. Serum glutamic oxaloacetic transaminase (SGOT)
 - b. Serum glutamic pyruvic transaminase (GPT)

Component 4. Performance of Darag Native Chicken (*Gallus gallus domesticus*) Given with Langkawas (*Alpinia galanga*) and Betel Nut (*Areca catechu*) Extract

Find out the productive performance of Darag native chicken in terms of:

- a. feed consumption
- b. gain in weight
- c. feed conversion ratio
- d. mortality and morbidity rate
- e. net profit

(11) Review of Literature/Prior Art Search

Darag native chicken are indigenous strains which are located only in Western Visayas. (WESVARDEC,2009). They are well known for their adaptability to local agro climatic conditions,hardiness, ability to utilize farm by products, tasty meat and a resistance to disease (PCAARD, 2007).

A parasite is an organism that lives in or on another organism (referred to as the host) and gains an advantage at the expense of that organism. The two types of internal parasites that affect poultry are worms and protozoa where round worms is the most common.

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Roundworms are common and are the largest worms affecting poultry (McKay, 2011). The large roundworm, *Ascaridia galli*, is one of the most common intestinal parasites found in chickens worldwide, especially free range and backyard flocks (<http://www.poultrydvm.com>). Chickens infected with *Ascaridia galli* worms suffer from slow growth and decreased weight gain (Prastowo and Ariyadi, 2019).

Protozoa are single-celled organisms found in most habitats, and they include some parasitic pathogens of humans and domestic animals. Protozoan parasites that are important to backyard poultry growers are coccidia (species of the *Eimeria* genus), cryptosporidia (*Cryptosporidium baileyi*), and histomonads (*H. meleagridis*). By far, the most common protozoan parasites of chickens and turkeys are coccidia (Jacob, 2019). Most commonly recognized coccidia infecting chickens and other poultry are of the genus *Eimeria* (Mc Dougald, 1998).

Galangal is widely used to treat breathing diseases, stomach diseases, diarrhea, and stomach cramps. Galangal can also function as an antimicrobial replacement for antibiotics. Galangal is also effective for treating fever, abnormal menstruation, and to increase male fertility. Galangal rhizome began to be used in several formulations to prevent cancer and tumors and is also used for the treatment of other diseases such as rheumatism, inflammation, diabetes, and neurological disorders. The galangal rhizome, has various compounds, namely essential oils, flavonoids, phenolic acids, saponins (Khairullah et al., 2020).

Betel nut, belongs to the family Arecaceae, which the seeds are used as an anthelmintic that contains alkaloids, arecoline and tannins. Arecoline is toxic to some worms and can paralyze them. Proanthocyanidins (a group of condensed tannins) can inhibit enzymes and degrade membranes. The inhibition of enzymes causes metabolic failure, which decreases energy production and leads to worm death. Tannins also act as anthelmintics through their ovicidal properties (Ademola et al., 2005).

The study relates to a prior art PH2/2012/000524 - Ethnobotanical Dewormer Composition for Native Chicken which presents a composition for an ethnobotanical dewormer for native chicken using the powder forms of mature seeds of betel nut (*Areca catechu*) and ipil-ipil (*Leucaena leucocephala*) as main ingredients. These plants contain constituents that serve as cheap sources of anthelmintics against internal parasitism. Internal parasitism in native chicken can cause severe diarrhea and high mortality. The formulation can be used for the treatment and control of roundworms, including common large roundworms of native chicken (*Ascaridia galli*), common thread worms (*Capillaria specie*), and the cecal worm (*Heterakis gallinarum*) and gape worm (*Syngamus trachea*). However, this prior art has different combination and dosages.

(12) Scientific Basis/Theoretical Framework

Study of Ozaraga et al., (2013) on powdered betel nut showed that 6 grams per kilogram body weight of native chicken can reduce intestinal parasites.

The result of the study of Yamson et al., (2019) revealed that after exposure of *Fasciola* spp. under 10%, 20%, and 40% extract concentrations, betel nut showed higher efficacy having the recorded mean motility time of 0.22, 0.07 min, and no movement upon contact, respectively, than Albendazole which produced mean motility time of 0.38 min.

(13) Methodology

Materials

The materials to be use in the study will includes the following:

One hundred eighty (180) heads of naturally infested Darag native chickens of both sexes, with ages ranging from 2 months to 4 months old; poultry house with rearing pens, serpentina plants, blender, knife, commercial feeds, weighing scale, feeders, waterers, old newspapers, electric microscope, test tubes, test tube racks, slides, Mc Master slide, sugar solutions, plastic bags, commercial coccidiostats and dewormer, , microtube, test tube with EDTA, centrifuge, camera, labeling materials and record book.

Methods

Experimental Design and Treatments

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Experimental Treatments. The study will be composed of two factors. Factor A will be the kind of plants extracts and commercial drugs and Factor B are the different dosages. Factor A includes: A1 – Control, A2 – Langkawas extract , A3 – Betel nut extract, A4 –Combined Plant Extract, A5- Commercial Dewormer and A6- Commercial Coccidiostat. Factor B will be: B1 – 1.0 ml/kg BW and B2 – 2 ml/kg BW of Darag native chickens.

Experimental Lay-out and Design. A 6x2 factorial experiment in Completely Randomized Design (CRD) will be used in the study. A total of 180 naturally infested experimental birds will be randomly distributed to twelve treatments, each replicated three times. Randomization will be done using the draw-lots scheme.

Cultural Management and Procedures

Preparation of Housing and Equipment. A total of 36 cages with a floor space area of 1 square meter each will be prepared to accommodate five experimental birds to represent one experimental unit. The cages, feeders and waterers will be cleaned with soap and water and disinfected using common commercial disinfectant available in the market. The cages and equipment will be allowed to air – dry for one week to ensure that the active ingredients will be evaporated. The cleanliness of the cages and other materials will also maintained.

Procurement and Quarantine of Experimental Animals. Experimental birds (pure Darag native chickens) will be taken from CapSU Darag Native Chicken Research Project. The birds will be quarantined for two weeks in the quarantine house away from the research site. They will also given with multivitamins and electrolytes to recover from stress during transport. After the quarantine period, they will be transferred to their assigned pens for data gathering.

Feeds and feeding. *Ad libitum* feeding will be provided with a grower ration (mash or crumble) in the duration of the study. Experimental birds will be fed twice daily; one in the morning and one in the afternoon.

Provision of water. The experimental birds will be provided with fresh and clean tap water every day. Waterers will be cleaned and replaced when found dirty. Antibiotic and vitamins will be given during the stressful conditions particularly before and after body weights will be taken and during blood collection.

Health Management of Experimental Birds. Prior to the start of the study, all experimental birds will be given with Newcastle disease vaccine to avoid disease outbreak. However, they will not given with any dewormer in preparation of the study. During stress days such as weighing, blood collection, and in rainy seasons, they will be given with anti stress supplements.

Preparation of Langkawas and Betel nut Plants Extract. The plants will be collected, washed, sundried and subject for ethanolic extraction.

Administration of Treatments. Ethanolic plant extract and commercial drugs will be weighed based on the dosage required and will be mixed to water. Treatments will be administered and will be repeated 14th day after the first administration.

Phyto chemical Evaluation. Langkawas and Betel nut plant extract will be brought to WVSU Central Laboratory for phytochemical analysis.

Data Gathering Procedures

Effects of Serpentina Plants in the Egg/ Oocyst per Gram (EPG/OPG) Count

The effect of serpentina plants will be evaluated based on the reduction of egg per gram count and coccidian oocyst count of the intestinal parasites which will be estimated by establishing egg per gram count (EPG) and oocyst per gram (OPG) count before and after administration of the plants extract and the commercial drugs. The data to be collected will be the egg and oocyst of the different species of intestinal parasites from the experimental birds. Pre - treatment egg/ oocyst per gram (EPG and OPG) count will be gathered for three consecutive days before the administration of plants extract. Post - treatment egg/ oocyst count will be done on 7th, 14th, 21st and 28th days after administration of plant extracts on the first month and follow up on the 2nd month.

Fecal Sample Collection. Fecal samples will be randomly collected from the mats that will be laid in the birds' cages early in the morning and will be brought to the College of Veterinary Medicine Diagnostic Laboratory for fecal examination. Quantitative and qualitative examinations will be used to determine the number of egg per gram (EPG) and oocyst per gram (OPG) count using the flotation

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technique and Mc Master Method. The average oocyst/ egg per gram count will be obtained by examining each sample thrice.

Hematological and Serum Biochemical Parameter Analysis. To determine the hematological value and serum biochemical parameters of experimental birds, pre treatment analysis will be done prior treatment administration and post treatment at 12 weeks after treatment. These blood values will includes hematological parameters such as total red blood cell count (RBC) and white blood cell count (WBC) and biochemical values in terms of Serum glutamic oxaloacetic transaminase (SGOT) and Serum glutamic pyruvic transaminase (SGPT). Blood samples will be withdrawn from the experimental birds and this will be examined in any Diagnostic laboratory (Biofera) for the hematological and serum biochemical test.

Productive Performance

Productive performance will be determined to measure the effect of plants extracts by determining the gain in weight, feed consumption, feed conversion ratio, mortality and morbidity rate and the net profit of birds (Ozaraga, 2012). Below are the procedures in data gathering of productive performance of experimental animals:

Gain in weight. The gain in weight (grams) will be determined by weighing the experimental birds at the start and at the end of the experimental period. The initial weight obtained will be subtracted from the final weight of the birds. Weighing of the birds will be done in the morning before feeding to minimize error on the 7th, 14th, 21st and 28th day, 2nd month and 3rd month post treatment. The gain in weight will be expressed in grams (Abella, 2013).

Feed consumption. The birds' feed intake (grams) will be determined daily. Feeds will be weighed using a 1000-gram capacity weighing scale. In the morning, left over and wastage will be collected and subtracted to the amount given the previous day. The difference represents the amount of feed consumed.

Mortality and morbidity rate. The health condition and status of birds will be strictly observed. Cases of birds' death and sickness will be noted.

Net Profit. The net profit will be calculated by subtracting the total expenses from the total sales obtained from the live chickens. This will be done by computing all expenses incurred in the study and will be subtracted from the total sales of the birds after 90 days of experimental period.

Data Analysis Procedures

The data to be gathered will be recorded, tabulated and analyzed according to the needs of the study. Percent reduction of intestinal parasites will be obtained using the formula:

$$\text{Percent Reduction} = \frac{A-B}{A} \times 100$$

Where:

A = Pre-treatment egg/oocyst per gram count

B = Post-treatment egg/oocyst per gram count

The statistical analyses of the data to be gathered for the various parameters tested will be analyzed and statistically interpreted using a 6x2 factorial in Completely Randomized Design (CRD).

Data collected will be analyzed using the Analysis of Variance (ANOVA). Differences in treatment means will be compared using the Duncan Multiple Range Test (DMRT). The result will be interpreted at five percent level of significance.

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(14) Major Activities/Work plan

RESEARCH ACTIVITIES	Jan	Feb	Mar	Apr	May	Jun	Jul-Dec
Acquisition of materials							
Data Gathering and Analysis							
Technical reports/presentations							
UM/ Patent Application							
IEC materials/Flyers production							

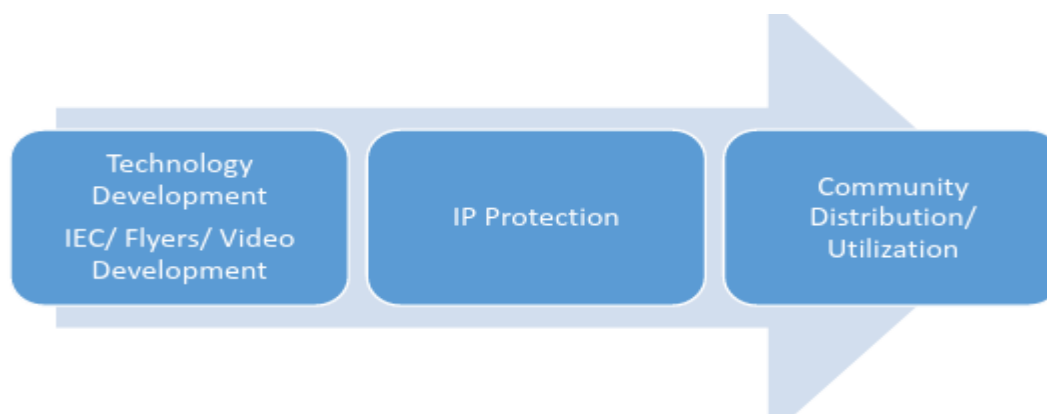
(15) Expected Output

EXPECTED OUTPUTS	DETAILS	TARGET
People Services	Train/ mentored native chicken growers and group or association for the utilization of langkawas and betel nut dewormer	100
Publications	Develop/ produce Information, Education and Communication (IEC) and promotional materials	2
Patents/IP	File 2 copyrights for the developed IEC materials and videos.	2
Places and Partnerships	Forge linkages and partnerships to LGU's, Capiz Darag Growers Association, Capiz Tourism Office, DTI, and other organizations.	5
Products	Langkawas and Betel nut Dewormer/ Coccidiostat	
Policy	Develop one policy inputs for utilization of langkawas and betel extract as dewormer fro chickens	1
Impact to Social/ Economics	<i>Social Impact:</i> - Control of parasitic infection to chicken - Technology transfer and commercialization strengthened - Strengthened Public-Private Partnerships - Innovation ecosystem improved - Technopreneurship empowered - Local agribusinesses supported	

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	<i>Economic Impact:</i> Increase income of native chicken, growers		
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(15) Technology Transfer Activity Design



(16) Targets Beneficiaries

- Chicken Raisers
- Department of Agriculture
- Bureau of Animal Industry

(17) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization
Dr. Maryneth B. Barrios	Study Leader	Animal Science
Dr. Leah A. Ingalla	Member	Animal Science
Dr. Joseph M. Lachica	Member	Animal Science
Research Assistant		
Laborer		

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (CapSU)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	2 @ 11,068 x 12 months	265,632
B. Maintenance and Other Operating Expenses (MOOE)		
Transportation allowance		30,000
Representation Expenses		30,000

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Training Expenses		40,000
Supplies (Feeds and Veterinary Medicines)		100,000
Communication		5,000
Laboratory Analysis		20,000
<i>C. Capital Outlay</i>		
Repair and Maintenance of DNC Housing		100,000
Total		592, 632.00

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